

PROJECT DOCUMENTATION

Project Title	Automated Network Request Management in ServiceNow
Student	Bandhanadam Antony
Project Type	Individual / Solo Project
Platform	ServiceNow Personal Developer Instance (PDI)
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This document records the table, automation, policy, and catalog configuration used to manage Network Requests through the ServiceNow Service Catalog and Flow Designer implementation.

1. Custom Table Configuration

Activity 1: Creation of Table

The foundation for the solution is the Network Database custom table. The following figures show the initial table creation and its basic configuration.

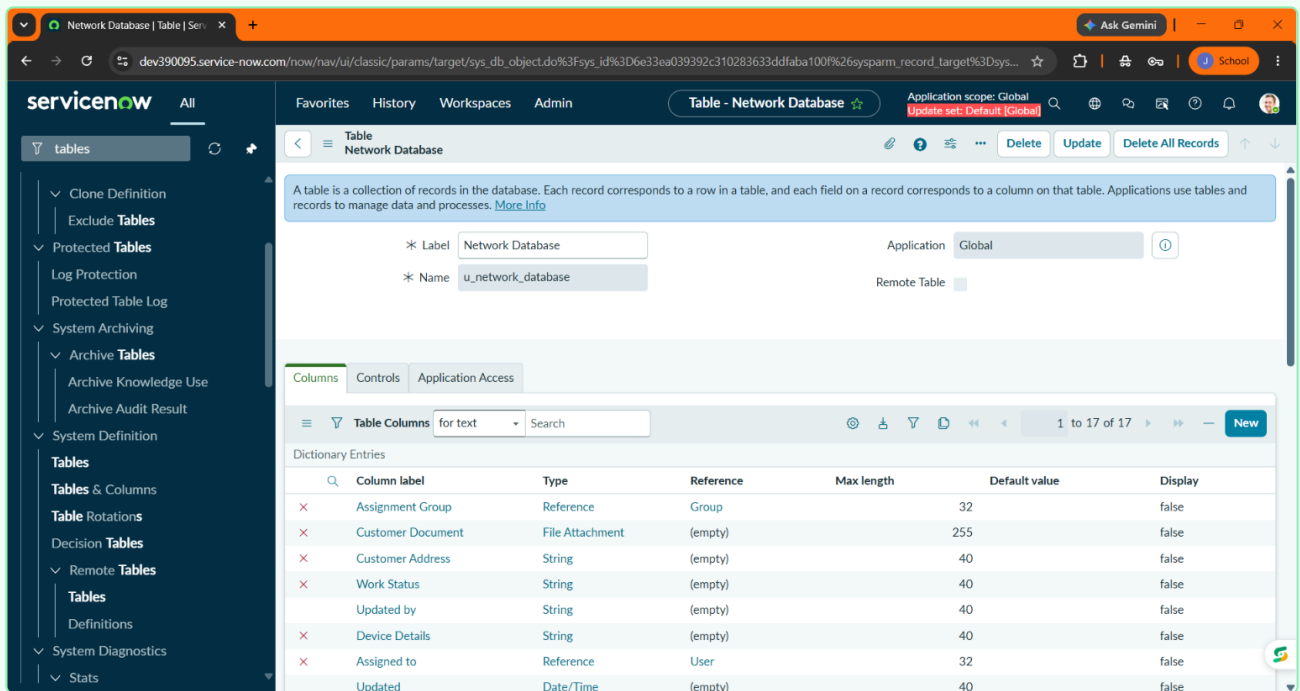


Figure 1. Network Database custom table.

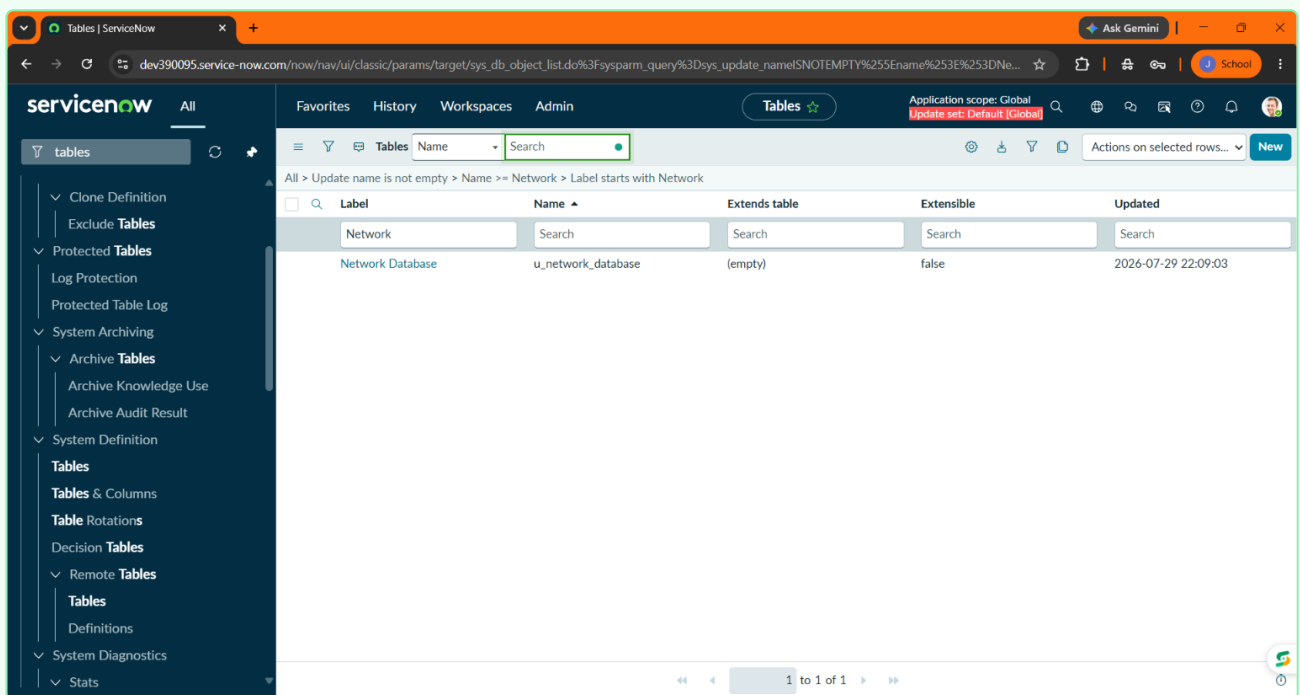


Figure 2. Network Database table configuration.

Activity 2 & 3: Creation of Fields and Defining Properties

We define the necessary fields required for the request records, applying the correct data types, max lengths, labels, and access controls for security.

Access Controls (4)

Security Data Filters

Labels (1)

Database Indexes (3)

Table Subscription Configuration (1)

Name

Search

Actions on selected rows...

Access Controls

☐	Name	Decision Type	Operation	Type	Active	Updated by	Updated
☐	u_network_database	Allow If	delete	record	true	admin	2026-07-29 22:09:04
☐	u_network_database	Allow If	write	record	true	admin	2026-07-29 22:09:04
☐	u_network_database	Allow If	read	record	true	admin	2026-07-29 22:09:04
☐	u_network_database	Allow If	create	record	true	admin	2026-07-29 22:09:04

<<

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1

to 4 of 4

>

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Figure 3. Columns and field properties.

FavoritesHistoryWorkspacesAdmin

Table - Network Database

Application scope: Global
Update set: Default (Global)

Table
Network Database

Table Columns

for text

Search

1 to 17 of 17

New

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
✖	Assignment Group	Reference	Group		32	false
✖	Customer Document	File Attachment	(empty)		255	false
✖	Customer Address	String	(empty)		40	false
✖	Work Status	String	(empty)		40	false
	Updated by	String	(empty)		40	false
✖	Device Details	String	(empty)		40	false
✖	Assigned to	Reference	User		32	false
	Updated	Date/Time	(empty)		40	false
✖	Type Of Devices	Choice	(empty)		255	false
	Sys ID	Sys ID (GUID)	(empty)		32	false
	Created by	String	(empty)		40	false
✖	Request For	String	(empty)		40	false
✖	Email	String	(empty)		40	false
✖	Request Number	String	(empty)		40	false
	Updates	Integer	(empty)		40	false
	Created	Date/Time	(empty)		40	false
✖	Date Of Enquiry	Date	(empty)		255	false
+	Insert a new row...					

Figure 4. Access controls for Network Database.

2. Flow Designer Automation

Activity 1 & 2: Creation of Flow and Configuring Trigger

The flow automates the lifecycle of a request. It begins by triggering upon catalog submission and retrieving the necessary variables provided by the requester.

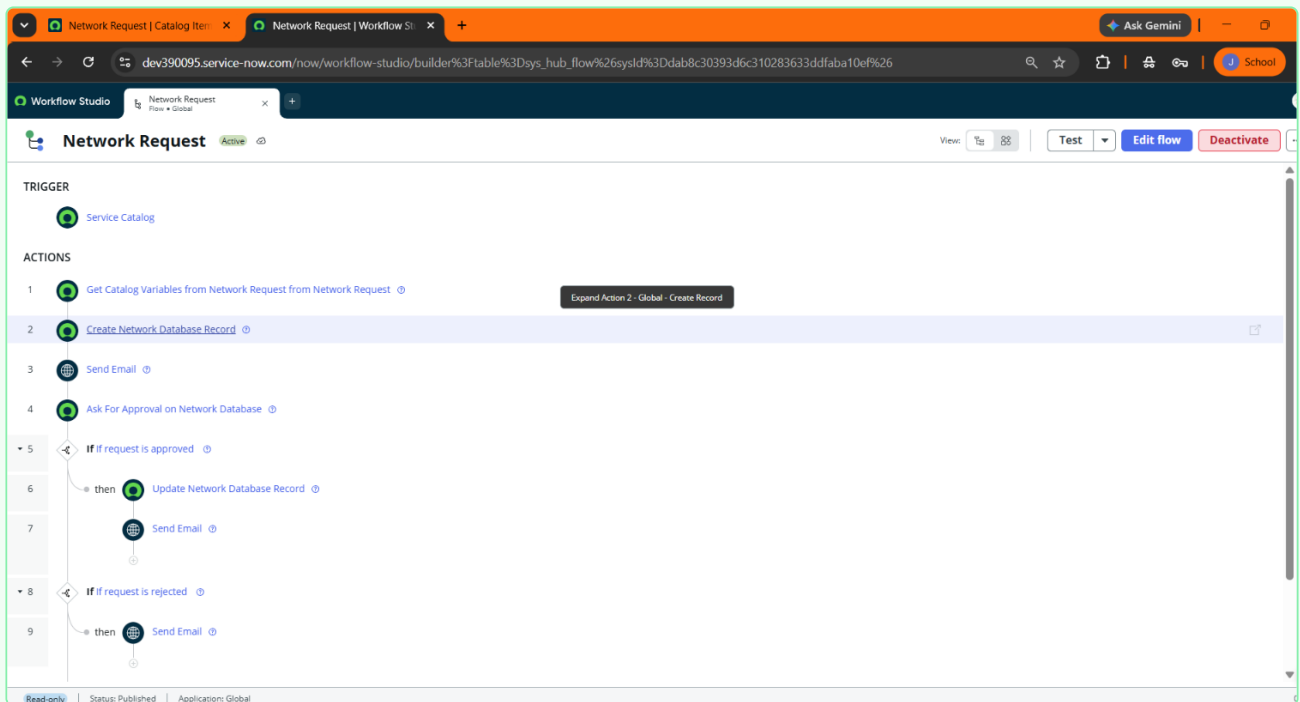


Figure 5. Network Request flow chart in Flow Designer.

Action Properties

Action: **Get Catalog Variables**

Action Inputs

* Submitted Request (Requested Item): Trigger: ... Requested Item ...

Select one or more values from the Template Catalog Items and Variable Sets, and select the required Catalog Variables to generate output data pills. You cannot choose the same Catalog Variable from multiple Template Catalog Items and Variable Sets.

* Template Catalog Items and Variable Sets (Catalog Items and Variable Sets): **Network Request**

Catalog Variables

Available: No available values

Selected: on_the_behalf_of, email, phone_number, user_name, proof_of_document, request_for, is_this_a_new_connection_c, if_this_is_a_relocation_plea, please_provide_address_he, type_of_devices

Note: If removing a variable from the 'Selected' list, it will be moved to 'Available' list only if the variable is from the selected Template Catalog Items and Variable Sets. Otherwise, the variable is removed from both 'Available' and 'Selected' lists.

Figure 6. Get Catalog Variables action.

Activity 3: Configuring Actions - Approval

An approval action is implemented followed by a decision condition to dynamically route the workflow based on whether the request is approved or rejected.

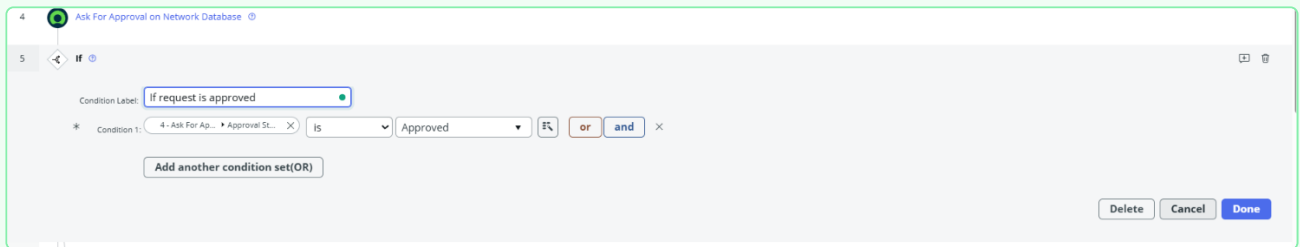


Figure 7. Ask for Approval action.

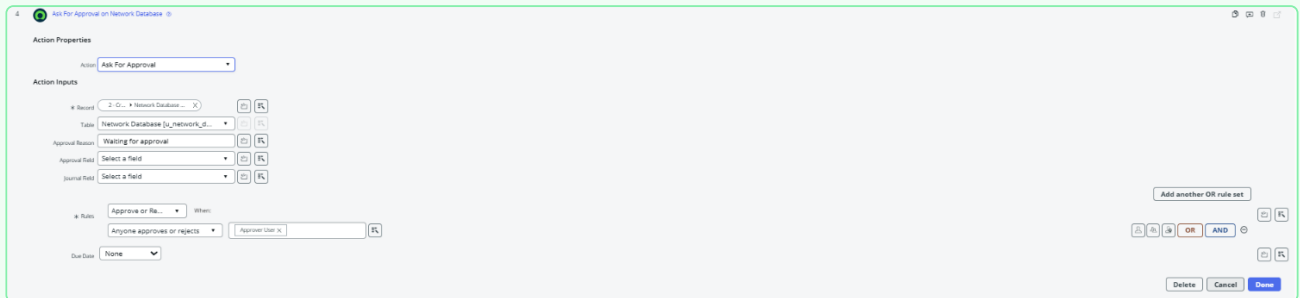


Figure 8. Approved decision condition.

Approved Request Path

For approved requests, the flow updates the current request status, creates a new entry in the Network Database using the catalog variables, and sends a confirmation email.

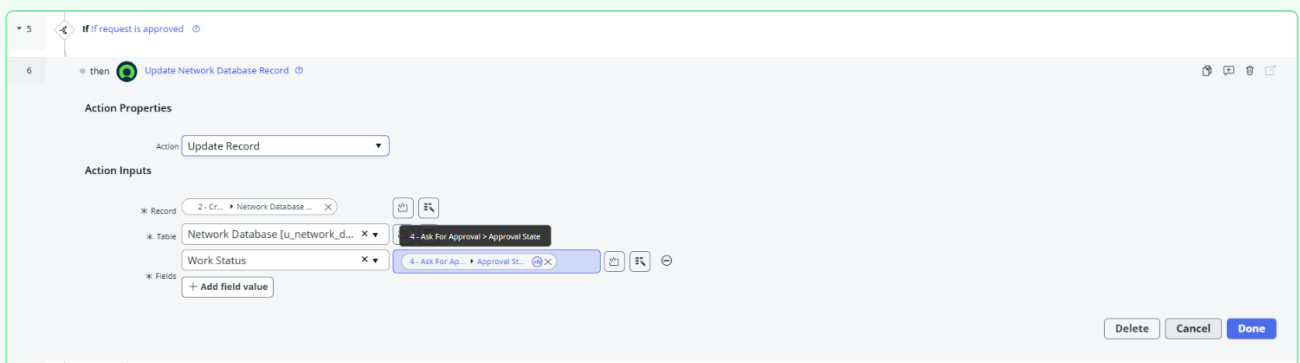


Figure 9. Approved-path Update Record action.

2  Create Network Database Record 

Action Properties

Action Create Record

Action Inputs

* Table  

Request Number	X ▼	Trigger - Servic... ▶ ... ▶ Num...	  
Request For	X ▼	1 - Get Catalog Va... ▶ ... ▶ Na...	  
Work Status	X ▼	New	  
Assignment Group	X ▼	Network X ▼	  
Date Of Enquiry	X ▼	1 - Get Catalog ... ▶ ... ▶ Creat...	  
Device Details	X ▼	1 - Get Catalo... ▶ type of dev...	  
* Fields			
Customer Address	X ▼	1 - Ge... ▶ provide device det... ▶ If this is a relocation pleas...	  
Type Of Devices	X ▼	1 - Get Catalo... ▶ type of dev...	  
Customer Document	X ▼	1 - Get Cat... ▶ proof of docu...	  
Email	X ▼	1 - Get Catalog Va... ▶ ... ▶ Em...	  
Assigned to	X ▼	1 - Get Catalog V... ▶ request...	  
+ Add field value			

Figure 10. Create Record action and field mapping.

3  Send Email 

Action Properties

Action Send Email

Action Inputs

Target Record 2 - Cl... ▶ Network Database...  

Table Network Database [u_network_d... 

Include Watermark ☒  

* To Trigger - Service C... ▶ ... ▶ Em...  

CC  

BCC  

* Subject Request has been created  



OpenGaisLo... 1 

Body

Hello! 1 - Get Catalog Va... ▶ ... ▶ Em...

we have been received your request! 2 - Crea... ▶ ... ▶ Request No...

sorry for the inconvenience and your request will resolved in 2 business working days.

Thankyou

Figure 11. Send Email action for approval.

Action Properties

Action: Send Email

Action Inputs

Target Record: 2 - Cr... Network Database ...

Table: Network Database [u_network_d...

Include Watermark: ☒

* To: 1 - Get Catalog Va... Em...

CC:

BCC:

* Subject: Request is approved

Body: Request is approved

Buttons: Delete, Cancel, Done

Figure 12. Approval email received by the requester.

Rejected Request Path

If the request is rejected, the flow records this outcome, updates the request record accordingly, and sends a notification email explaining the rejection.

Action Properties

Action: Update Record

Action Inputs

* Record: 2 - Cr... Network Database ...

* Table: Network Database [u_network_d... X

* Fields: Work Status X Rejected

+ Add field value

Buttons: Delete, Cancel, Done

+ Add an Action, Flow Logic, or Subflow

Figure 13. Rejected decision condition.

Condition Label: If request is rejected

* Condition 1: 4 - Ask For Ap... Approval St... is Rejected

+ Add another condition set(OR)

Buttons: Delete, Cancel, Done

Figure 14. Rejected-path Update Record action.

9 • then Send Email

Action Properties

Action: Send Email

Action Inputs

Target Record: 2 - Cr... Network Database ...

Table: Network Database [u_network_d...

Include Watermark: ☒

* To: 1 - Get Catalog Variabl... em...

CC:

BCC:

* Subject: Request is rejected

Body: Request is rejected

Formatting: OpenSans.So... 10pt

Delete Cancel Done

Figure 15. Rejection email action.

3. Catalog UI Policy and Client Script

To ensure data accuracy and guided form behaviour, Catalog UI Policies and a Client Script are applied. These dictate field visibility and mandatory requirements dynamically.

Variables (12)

Variable Sets (1)

Catalog UI Policies (3)

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

≡

Order

▼

Search

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Figure 16. Network Request Catalog UI policies.

Variables (5)

Catalog UI Policies

Catalog Client Scripts (1)

Included In (1)

Catalog Data Lookup Definitions

≡

for text

Search

🔍

🔗

🔍

📄

—

Actions on selected rows...

New

Variable set = Request information

☐	☐	Name	Active	Table	View	Type	Updated
		Requester details	true			onChange	2026-07-30 00:49:56

⏪

⏴

1

to 1 of 1

⏵

⏩

Figure 17. Catalog UI policy variable actions.


```

1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || !newValue) {
3     g_form.setValue('email', '');
4     g_form.setValue('user_name', '');
5     g_form.setValue('phone_number', '');
6     return;
7   }
8
9   var gr = g_form.getReference('on_the_behalf_of', function(user) {
10    g_form.setValue('email', user.email);
11    g_form.setValue('user_name', user.user_name);
12
13    // Use Mobile Phone first, otherwise fall back to Business Phone
14    var userPhone = user.mobile_phone || user.phone || '';
15    g_form.setValue('phone_number', userPhone);
16  });
17 }

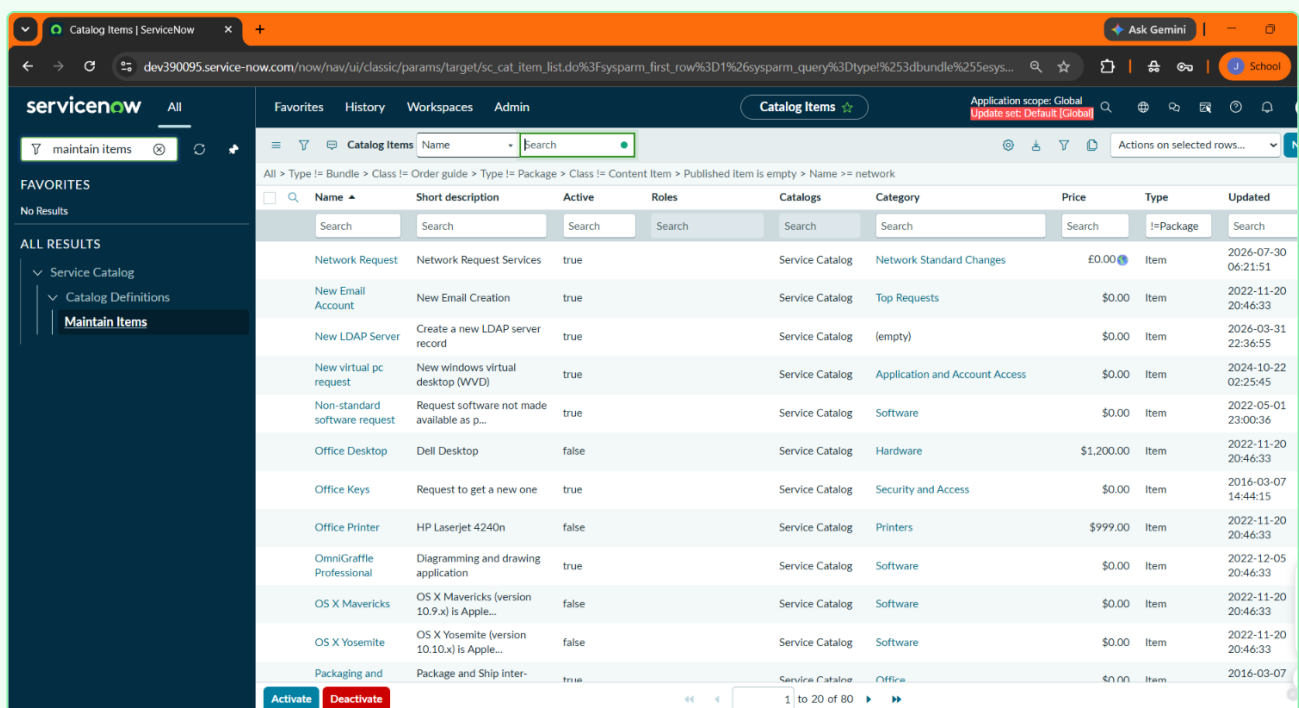
```

Figure 18. Catalog Client Script.

4. Service Catalog Configuration

Activity 1: Creation of Service Catalog Item

The Service Catalog item is created within Maintain Items, categorized correctly, and configured for user visibility.



Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Network Request	Network Request Services	true		Service Catalog	Network Standard Changes	£0.00	Item	2026-07-30 06:21:51
New Email Account	New Email Creation	true		Service Catalog	Top Requests	\$0.00	Item	2022-11-20 20:46:33
New LDAP Server	Create a new LDAP server record	true		Service Catalog	(empty)	\$0.00	Item	2026-03-31 22:36:55
New virtual pc request	New windows virtual desktop (WVD)	true		Service Catalog	Application and Account Access	\$0.00	Item	2024-10-22 02:25:45
Non-standard software request	Request software not made available as p...	true		Service Catalog	Software	\$0.00	Item	2022-05-01 23:00:36
Office Desktop	Dell Desktop	false		Service Catalog	Hardware	\$1,200.00	Item	2022-11-20 20:46:33
Office Keys	Request to get a new one	true		Service Catalog	Security and Access	\$0.00	Item	2016-03-07 14:44:15
Office Printer	HP Laserjet 4240n	false		Service Catalog	Printers	\$999.00	Item	2022-11-20 20:46:33
OmniGraffle Professional	Diagramming and drawing application	true		Service Catalog	Software	\$0.00	Item	2022-12-05 20:46:33
OS X Mavericks	OS X Mavericks (version 10.9.x) is Apple...	false		Service Catalog	Software	\$0.00	Item	2022-11-20 20:46:33
OS X Yosemite	OS X Yosemite (version 10.10.x) is Apple...	false		Service Catalog	Software	\$0.00	Item	2022-11-20 20:46:33
Packaging and	Package and Ship Inter-	true		Service Catalog	Office	\$0.00	Item	2016-03-07

Figure 19. Maintain Items list.

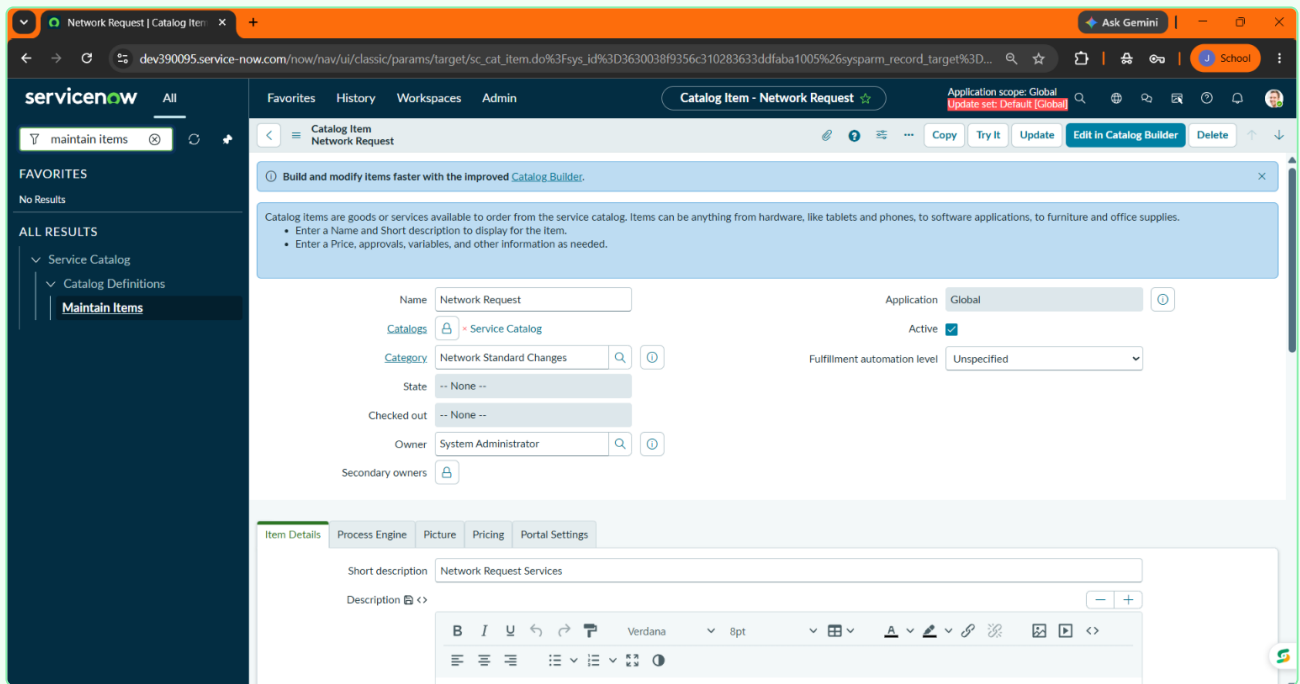


Figure 20. Network Request catalog item.

Activity 2 & 3: Variables Configuration and Types

The necessary questions are defined as catalog variables with specific types, ordering, and question text to gather the appropriate data from the user.

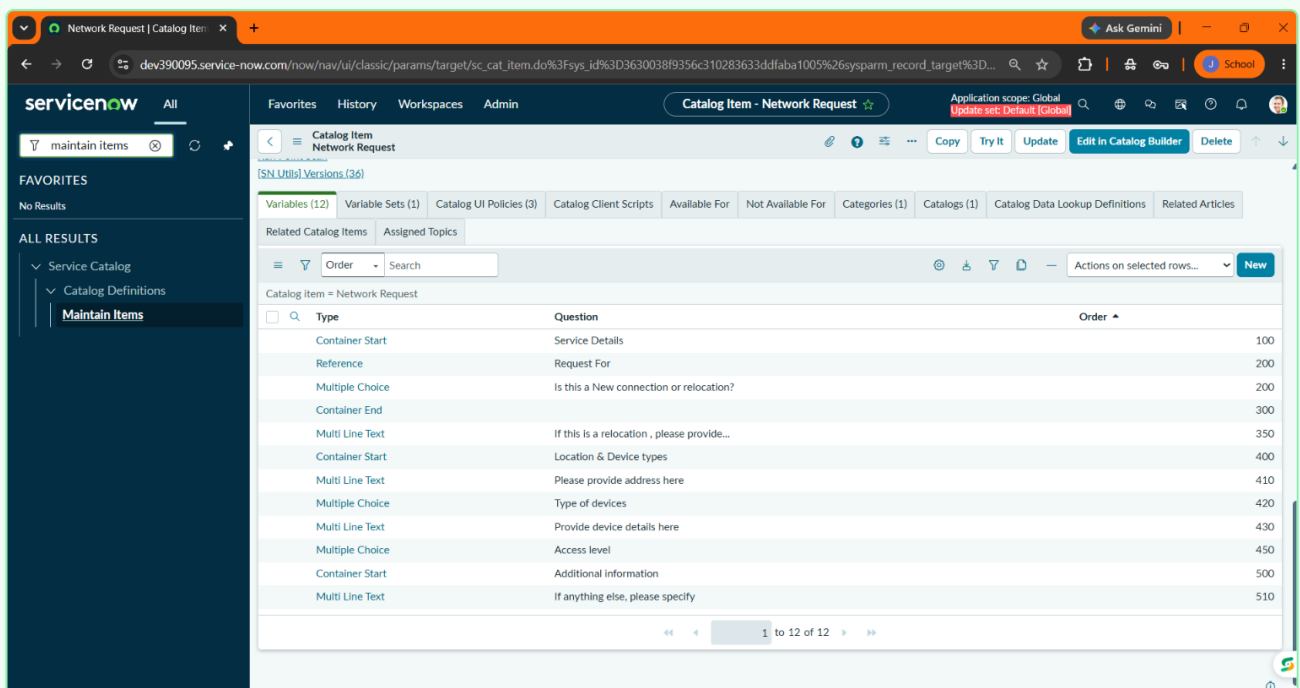


Figure 21. Catalog variables configured for Network Request.

Name	Type	Question	Order
on_the_behalf_of	Reference	on the behalf of	
email	Single Line Text	email	
phone_number	Single Line Text	phone number	
user_name	Single Line Text	user name	
proof_of_document	Attachment	proof of document	

Figure 22. Variable types and order.

Activity 4: Variable Set Configuration

A reusable variable set is created for consistency and attached to the main catalog item.

Variable set	Order
Request information	100

Figure 23. Network Request variable set.

Variable Set - Request Information

* Title: Request Information

* Internal name: request_information

Order: 100

Type: Single Row

Description:

Application: Global

Display title: ☒

Layout: 2 Columns Wide, one side, then the other

Figure 24. Variable set attached to the catalog item.

5. Catalog Submission & Validation

Activity 1 & 2: Search, Open, and Submit

Users can search for the catalog item, fill in the required fields guided by the UI policies, and submit their network request. A summary screen confirms the action.

The screenshot shows the 'Network Request' catalog item form in ServiceNow. The breadcrumb trail is: Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request. The form is titled 'Network Request' with a subtitle 'Network Request Services'. It includes a 'Request information' section with fields for 'on the behalf of' (Rachel Larrison), 'email' (rachel.larrison@example.com), 'phone number' (9645789521), 'user name' (rachel.larrison), and 'proof of document' (with an 'Upload' button). There is also a 'Request For' dropdown (Rachel Larrison) and a radio button selection for 'Is this a New connection or relocation?' (New is selected). A note at the bottom says '* Please provide address here' with a 'Guntur' input field. On the right, a sidebar shows 'Quantity: 1', 'Delivery Times: 2 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'.

Figure 25. Search results showing the Network Request catalog item.

The screenshot shows the ServiceNow search results for 'network request'. The breadcrumb trail is: Home > Search. The search bar contains 'network request'. On the left, there are filters for 'Sources' (All, Knowledge Bases, Catalogs) and 'FILTERS' (Knowledge Bases, Category, Author, Last modified, View Count) and 'Catalogs' (Category). The main results area, titled 'All results for "network request"', lists three items: 'Network Request' (with a link to 'Network Request Services'), 'Request for Backup' (with description 'Backup system, data files or database'), and 'Apple iPhone 13' (with description 'Request for Apple iPhone 13' and price '\$799.00'). Below it, 'Apple iPhone 13 pro' is listed with description 'Request for Apple iPhone 13 pro' and price '\$999.00 + \$45.00 Monthly'.

Figure 26. Network Request catalog form - requester information.

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Request Summary - REQ0010009 Search Catalog

Submitted : 2026-07-31 09:18:45
Request Number : REQ0010009
Requested for : Rachel Larrison
Estimated Delivery : 2026-08-02

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Network Request	2026-08-02	Assess or Scope Task	---	1	---

Total: \$0.00

Figure 27. Completed Network Request catalog form.

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request Search Catalog

* Please provide address here
Guntur

Type of devices
☐ Others
☒ Laptop
☐ Mobile

Provide device details here
Dell laptop

Access level
☐ High
☒ Low
☐ Medium

If anything else, please specify
Please solve as soon as possible

Add attachments
[Choose a file](#) or drag it here.
 Copy and paste clipboard files here.

Quantity: 1
 Delivery Time: 2 Days
 Add to Cart
 Save as Draft
 Order Now

Figure 28. Submitted Network Request summary.

6. Flow Output Verification

Activity 3: Verify Approved Record

Following approval, the system administrator verifies that the Flow Designer successfully created the corresponding record in the Network Database table with all correct mappings.

servicenow All Favorites History Admin : Network Database - Created 2026-07-31 21:22:30 Search

Network Database
Created 2026-07-31 21:22:30

Request Number: REQ0010009
Request For: Rachel Larrison
Work Status: approved
Customer Address: Dell laptop
* Type Of Devices: Laptop
Device Details: laptop

Email: rachel.larrison@example.com
Assigned to: Rachel Larrison
Date Of Enquiry: 2012-02-17
Customer Document: Click to add...
Assignment Group: Network

Update Delete

Figure 29. Network Database records created by the flow.

servicenow All Favorites History Workspaces Admin Network Databases Search

Network Databases Request Number Search Actions on selected rows... New

Request Number	Email	Device Details	Created	Date Of Enquiry
REQ0010008	annie.approver@example.com	laptop	2026-07-30 07:32:25	2011-09-16
REQ0010006	alene.rabeck@example.com	laptop	2026-07-30 06:26:43	2012-02-17
REQ0010007	allan.schwandt@example.com	mobile	2026-07-30 06:41:11	2012-02-17
REQ0010005	alejandro.mascall@example.com	laptop	2026-07-30 06:07:01	2012-02-17
REQ0010004	aileen.mottern@example.com	mobile	2026-07-30 05:52:49	2012-02-17
REQ0010009	rachel.larrison@example.com	laptop	2026-07-31 21:22:30	2012-02-17
REQ0010003	adela.cervantsz@example.com	mobile	2026-07-30 05:47:19	2012-02-17
REQ0010002	abraham.lincoln@example.com	laptop	2026-07-30 05:34:58	2013-07-23

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Figure 30. Approved Network Database request record.